Statistical Analysis of Market Data

Business Problem Investigated:

Q1. Test if the average income of 'Australia', 'India' and 'Canada' is same?

Q2. Test whether wine purchase across Australia, India and Canada has the same average ?

Q3. Is there association between Marital Situation and Income?

Q4. Is age group and number of Web visits per month independent?

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About the Data

Column Details:

ID: Customer's Unique Identifier
Year_Birth: Customer's Birth Year
Education: Customer's education level
Marital_Status: Customer's marital status
Income: Customer's yearly household income
Kidhome: Number of children in customer's household
Teenhome: Number of teenagers in customer's household
Dt_Customer: Date of customer's enrollment with the company
Recency: Number of days since customer's last purchase
MntFruits: Amount spent on fruits in the last 2 years
MntMeatProducts: Amount spent on meat in the last 2 years
MntFishProducts: Amount spent on fish in the last 2 years
MntSweetProducts: Amount spent on sweets in the last 2 years
MntGoldProds: Amount spent on gold in the last 2 years
NumDealsPurchases: Number of purchases made with a discount
NumWebPurchases: Number of purchases made through the company's web site
NumCatalogPurchases: Number of purchases made using a catalogue
NumStorePurchases: Number of purchases made directly in stores
NumWebVisitsMonth: Number of visits to company's web site in the last month
AcceptedCmp1: 1 if customer accepted the offer in the 1st campaign, 0 otherwise (Target variable)
AcceptedCmp2: 1 if customer accepted the offer in the 2nd campaign, 0 otherwise (Target variable)
AcceptedCmp3: 1 if customer accepted the offer in the 3rd campaign, 0 otherwise (Target variable)
AcceptedCmp4: 1 if customer accepted the offer in the 4th campaign, 0 otherwise (Target variable)
AcceptedCmp5: 1 if customer accepted the offer in the 5th campaign, 0 otherwise (Target variable)
Response: 1 if customer accepted the offer in the last campaign, 0 otherwise (Target variable)
Complain: 1 if customer complained in the last 2 years, 0 otherwise
Country: Customer's location

Data Preparation and Cleaning

- Review data.
- Check for duplicates and remove if any.
- Feature Engineering
- Look for missing values and perform imputation.

Observation:

- There isn't duplicate values.
- Income has null values.
- All are int and float values.
- "Income" has a space before it's name and $ symbol which can be problematic.

Remove $ and change the datatype to "int".

```
df.rename(columns={' Income ':'Income'},inplace=True)
df["Income"] = df["Income"].str.replace("$","").str.replace(',', '')
df["Income"] = df["Income"].astype(float)
```

Create variables for indepth analysis.

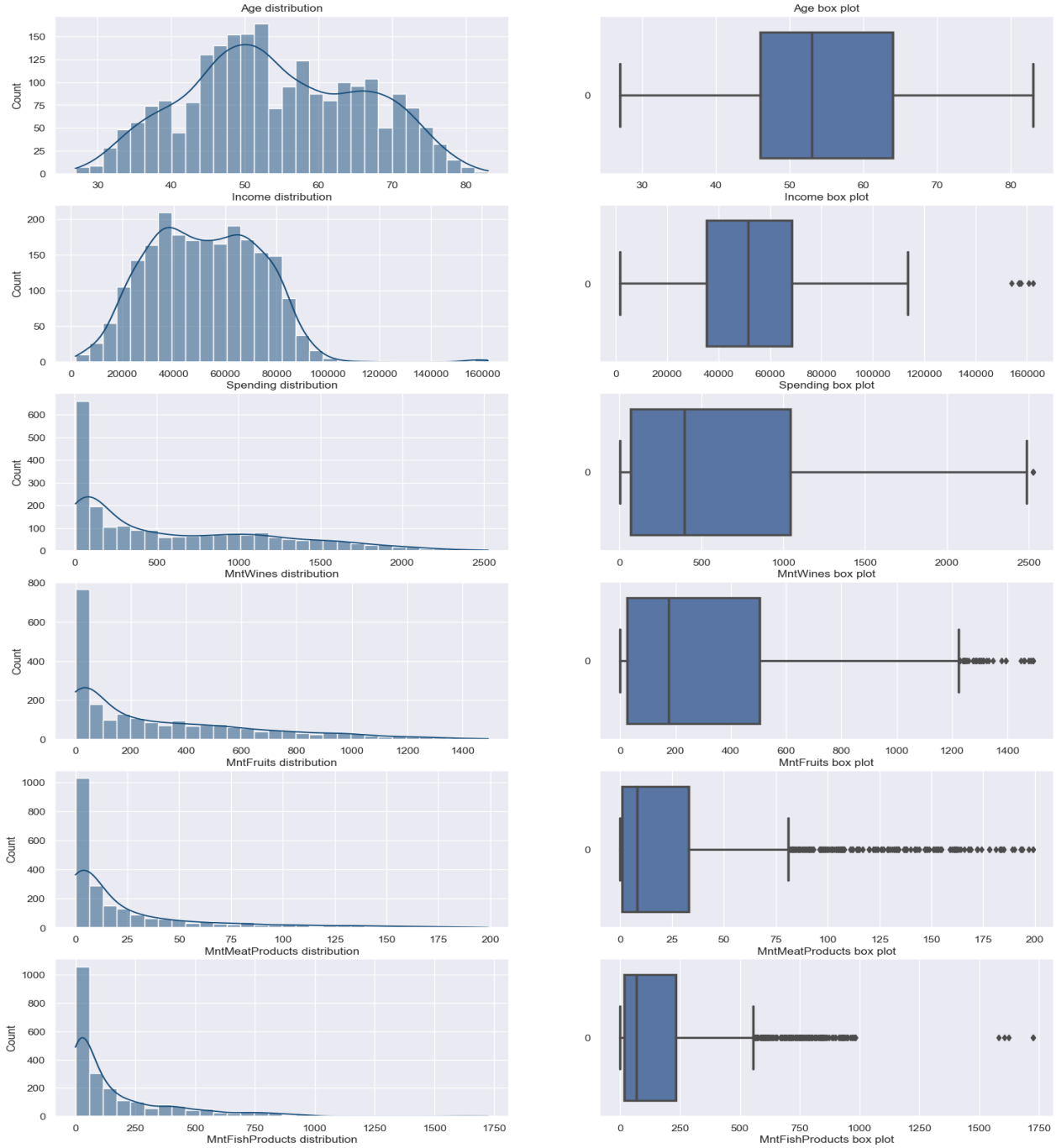
- "Age" is created based on birth year.
- "Spending" refers to sum of all individual spendings.
- "Marital_Situation" refers to grouping of marital status into single and in couple.
- "Has_child" is created after combining teenhome and kidhome columns.
- "Educational Years" represents approximate years spent on studies.

```
df['Age']=2023-df['Year_Birth']
df['Spending']=df['MntWines']+df['MntFruits']+df['MntMeatProducts']+df['MntFishProducts']
                +df['MntSweetProducts']+df['MntGoldProds']
df['Marital_Situation']=df['Marital_Status'].replace({'Divorced':'Alone','Single':'Alone','Married':'In couple',
                                                      'Together':'In couple','Absurd':'Alone','Widow':'Alone','YOLO':'Alone'})
df['Has_child'] = np.where(df.Kidhome+df.Teenhome > 0, 1, 0)
df['Educational_years']=df['Education'].replace({'Basic':5,'2n Cycle':8,'Graduation':12,'Master':18,'PhD':21})
```

```
bins = [0,20,30,40,50,60,70,100]
labels = ['Teen', 'Young', 'Young Adult', 'Adult', 'Senior', 'Retired', 'Old Age']
df['AgeGroup'] = pd.cut(df['Age'], bins=bins, labels=labels, right=False)
```

Test for Numerical Attributes

Visual Normality Checks:



Outlier Analysis

- Analyze the outliers using Tukey formula.
- After observing the histogram, we will remove outliers from age, income.

```
def tukey_outliers(x):
    q1 = np.percentile(x,25)
    q3 = np.percentile(x,75)


    iqr = q3-q1

    min_range = q1 - iqr*1.5
    max_range = q3 + iqr*1.5


    outliers = x[(x<min_range) | (x>max_range)]
    return outliers
```

```
df.iloc[list(tukey_outliers(df['Age']).index)]
```

MntFruits	MntMeatProducts	...	AcceptedCmp1	AcceptedCmp2	Response	Complain	Country	Age	Spending	Marital_Situation	Has_child	Educational_years
0	5	...	0	0	0	0	SA	130	22	Alone	1	8
144	562	...	0	0	0	0	SP	124	1853	In couple	0	21
6	8	...	0	0	0	1	IND	123	65	Alone	1	8

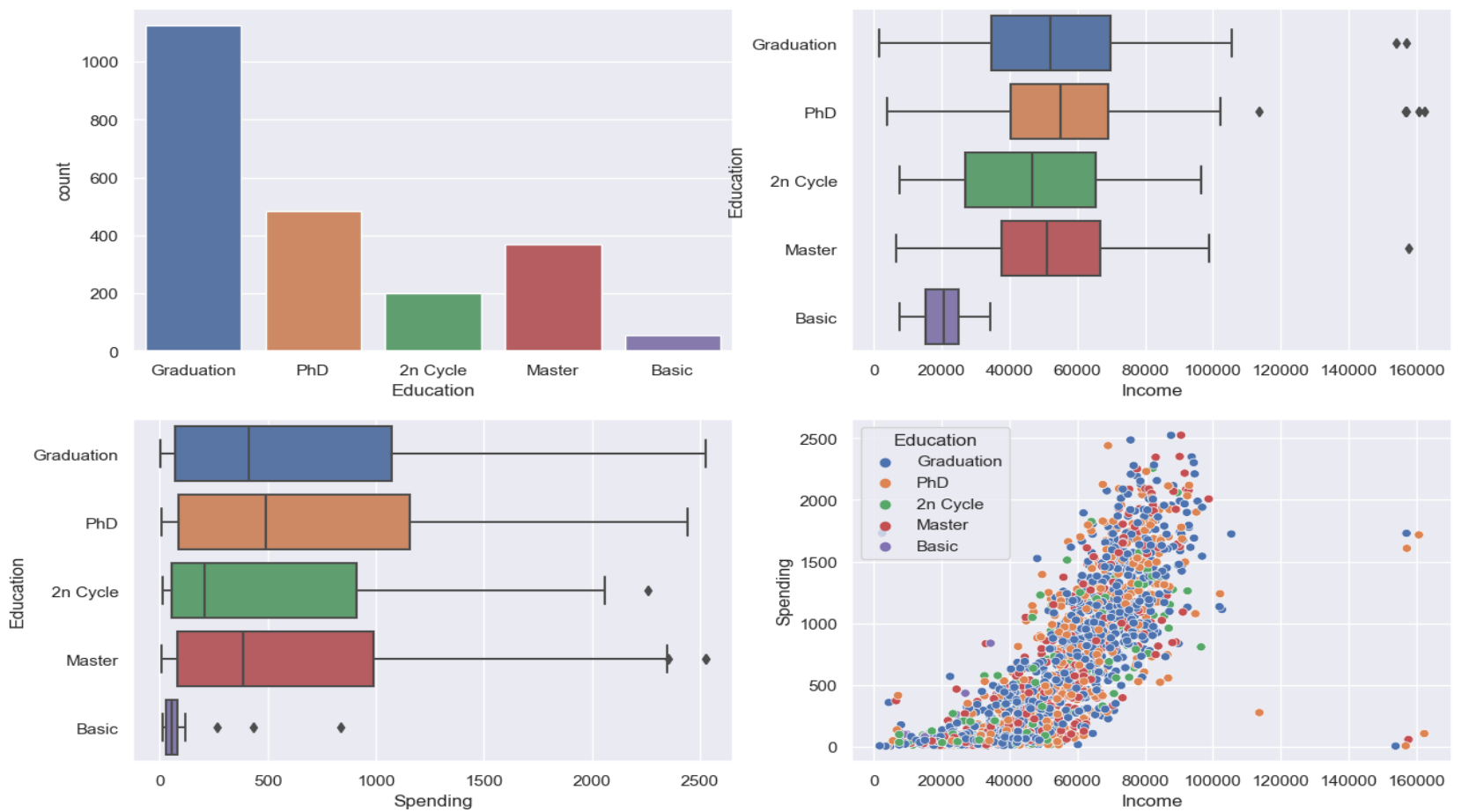
```
df = df.drop(tukey_outliers(df['Age']).index).reset_index(drop=True)
```

```
df = df.drop(df[df['Income'] >600000]['Income'].index).reset_index(drop=True)
```

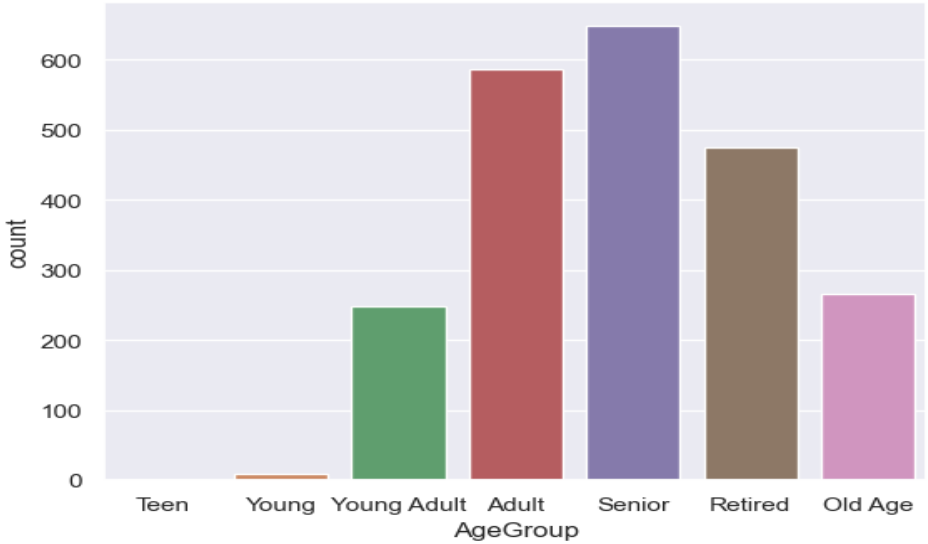
Multiple Imputation of Income

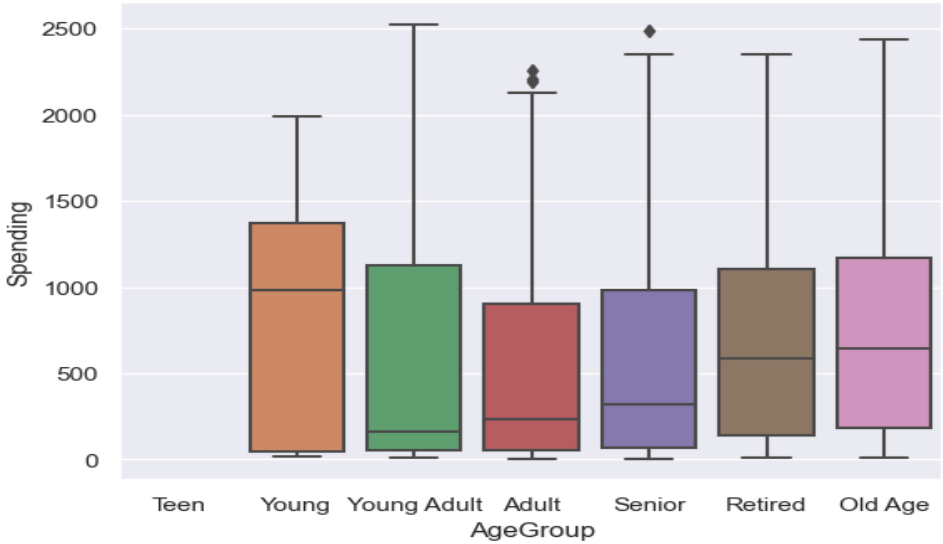
- The income though appears to be normal, is sensitive to Shapiro-Wilk test.
- After experimentation, income is imputed with multiple imputations.

Exploring Income



- We observe different trends with respect to income.
- Income is correlated with spending.



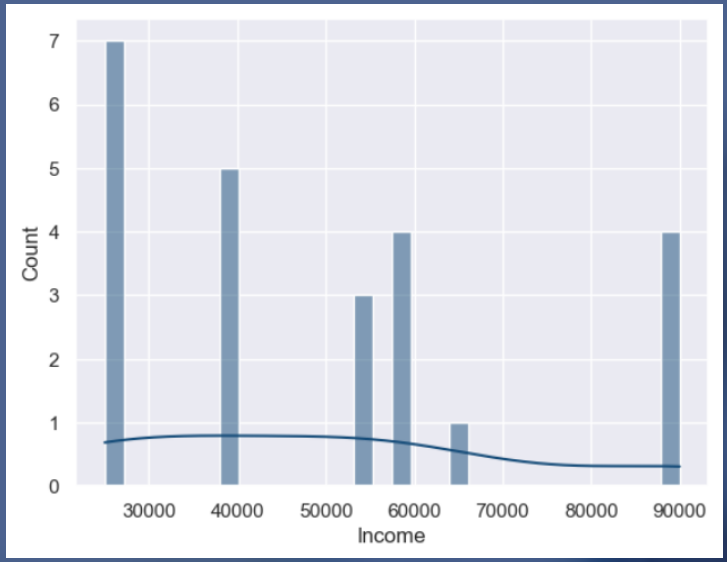


Examine spending of data with null Income

```
n_i = df[df['Income'].isnull()]
```

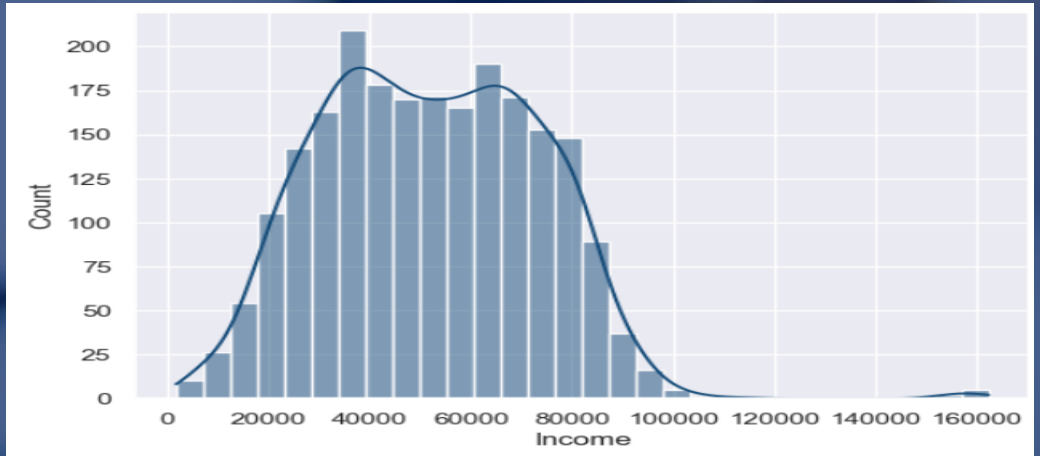
```
n_i[['Income', 'Spending']].describe()
```

	Income	Spending
count	0.0	24.000000
mean	NaN	487.875000
std	NaN	536.869291
min	NaN	18.000000
25%	NaN	46.750000
50%	NaN	262.000000
75%	NaN	722.000000
max	NaN	1679.000000

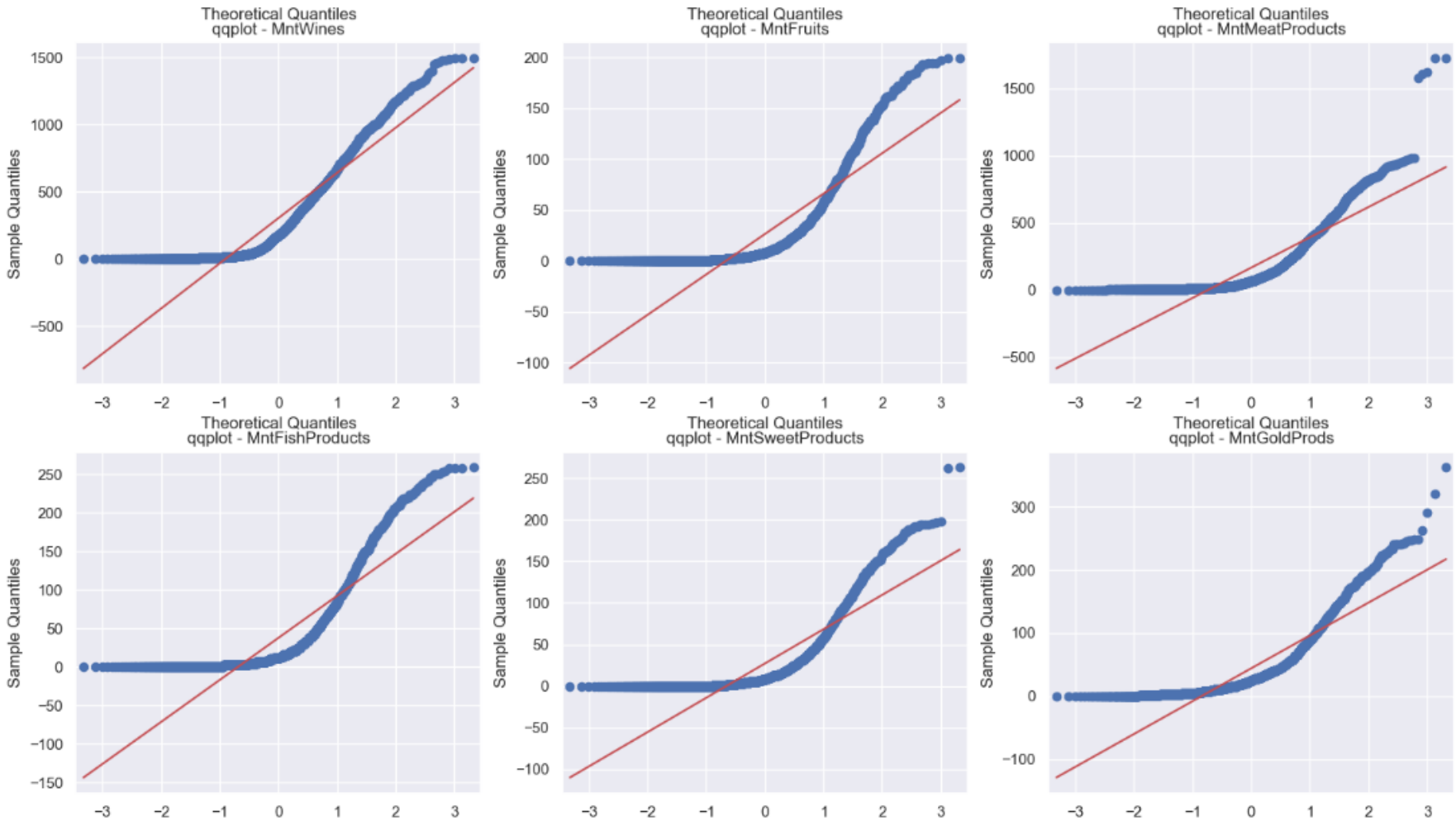


Apply Filters and assign Income

```
# Filters
c1 = (n_i['Spending'] >= 0) & (n_i['Spending'] <= 69)
c2 = (n_i['Spending'] > 69) &  (n_i['Spending'] <= 250)
c3 = (n_i['Spending'] > 250) & (n_i['Spending'] <= 500)
c4 = (n_i['Spending'] > 500) & (n_i['Spending'] <= 750)
c5 = (n_i['Spending'] > 750) & (n_i['Spending'] <= 1000)
c6 = (n_i['Spending'] > 1000) & (n_i['Spending'] <= 1750)

n_i.loc[c1, 'Income'] = 20000
n_i.loc[c2, 'Income'] = 40000
n_i.loc[c3, 'Income'] = 55000
n_i.loc[c4, 'Income'] = 58000
n_i.loc[c5, 'Income'] = 70000
n_i.loc[c6, 'Income'] = 100000
```



Quantile-Quantile Plot



- NumWebPurchases, NumCatelogPurchases, NumStorePurchases, NumWebVisitsMonth appears to be normal.
- Age though appears to be normal did have bimodal tendencies in the histogram plotted.

Examine normality for log transformed columns.

```
cols_log = ['Age',
 'Spending',
 'MntWines',
 'MntFruits',
 'MntMeatProducts',
 'MntFishProducts',
 'MntSweetProducts',
 'MntGoldProds',
 'NumDealsPurchases',
 'NumWebPurchases',
 'NumCatalogPurchases',
 'NumStorePurchases',
 'NumWebVisitsMonth']
```

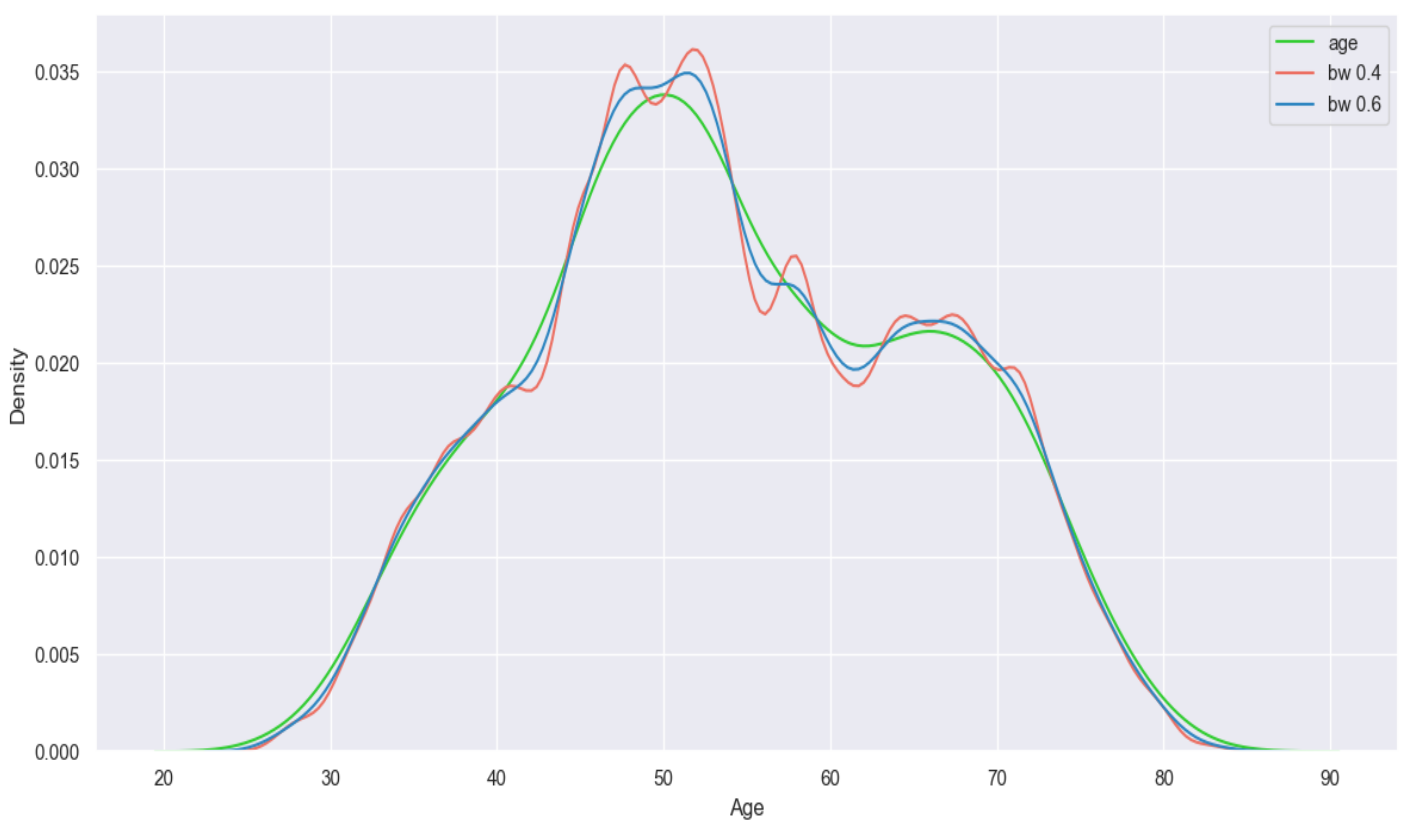
```
with np.errstate(divide='ignore'):
    df_log = pd.DataFrame(np.log2(df[cols_log]))
```

```
shapiro_wilk_test(df_log, cols_log)
```

```
Rejected H0 under significance level 0.05
Age doesn't seems to be normally distributed

Rejected H0 under significance level 0.05
Spending doesn't seems to be normally distributed

Fail to reject H0 due to lack of evidence under significance level 0.05
MntWines seem to be normally distributed

Fail to reject H0 due to lack of evidence under significance level 0.05
MntFruits seem to be normally distributed

Fail to reject H0 due to lack of evidence under significance level 0.05
MntMeatProducts seem to be normally distributed

Fail to reject H0 due to lack of evidence under significance level 0.05
MntFishProducts seem to be normally distributed

Fail to reject H0 due to lack of evidence under significance level 0.05
MntSweetProducts seem to be normally distributed

Fail to reject H0 due to lack of evidence under significance level 0.05
MntGoldProds seem to be normally distributed

Fail to reject H0 due to lack of evidence under significance level 0.05
NumDealsPurchases seem to be normally distributed

Fail to reject H0 due to lack of evidence under significance level 0.05
NumWebPurchases seem to be normally distributed

Fail to reject H0 due to lack of evidence under significance level 0.05
NumCatalogPurchases seem to be normally distributed

Fail to reject H0 due to lack of evidence under significance level 0.05
NumStorePurchases seem to be normally distributed

Fail to reject H0 due to lack of evidence under significance level 0.05
NumWebVisitsMonth seem to be normally distributed
```

Bimodal Data

```
from sklearn.neighbors import KernelDensity

age = df['Age'].to_numpy().reshape(-1, 1)

kd = KernelDensity(kernel='gaussian', bandwidth=0.2).fit(age)
```



Test if Income of 'Australia', 'India' and 'Canada' has same mean?

Check for same Mean in 2 samples

- If the mean accurately represents the center of your distribution and your sample size is large enough, consider a parametric test because they are more powerful.
- If the median better represents the center of your distribution, consider the nonparametric test even when you have a large sample.
- Age, income, NumWebPurchases has mean and median similar representing center of the distribution. Thus parametric tests can be used.
- Non parametric test can be applied on other columns.

```
mm_df = pd.DataFrame()
mm_df['mean'] = df[cols].mean()
mm_df['median'] = df[cols].median()
mm_df['diff'] = (df[cols].mean() - df[cols].median())/df[cols].mean() * 100
```

	mean	median	diff
Age	54.101968	53.0	2.036835
Income	51958.810579	51371.0	1.131301
Spending	605.986583	396.5	34.569508
MntWines	304.127460	174.0	42.787146
MntFruits	26.275939	8.0	69.553895
MntMeatProducts	166.983453	67.0	59.876264
MntFishProducts	37.536225	12.0	68.030883
MntSweetProducts	27.080501	8.0	70.458449
MntGoldProds	43.983005	24.0	45.433470
NumDealsPurchases	2.326029	2.0	14.016535
NumWebPurchases	4.087657	4.0	2.144420
NumCatalogPurchases	2.663238	2.0	24.903442
NumStorePurchases	5.795617	5.0	13.727911
NumWebVisitsMonth	5.318873	6.0	-12.805852

Non Parametric Tests.

Non-parametric test (sometimes referred to as a distribution free test) does not assume anything about the underlying distribution (for example, that the data comes from a normal (parametric distribution).

Advantages:

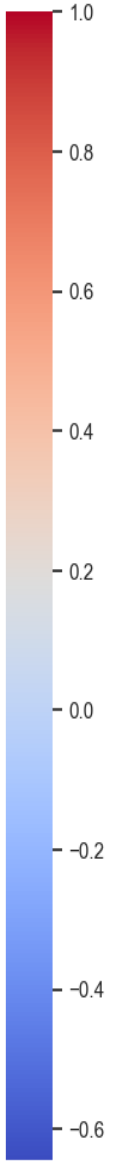
- More statistical power when assumptions of parametric tests are violated.
- Assumption of normality does not apply.
- Small sample sizes are okay.
- They can be used for all data types, including ordinal, nominal and interval (continuous).
- Can be used with data that has outliers.

Pearson Correlation

- The Pearson correlation coefficient is just a normalized covariance between the two variables to give an interpretable score such that Corr p(x,y)∈[−1,1]
- It can be used to summarize the strength of the linear relationship between two data samples.

- We observe correlation between 'Spending' and 'Income'.
- 'MntWines', 'MntMeatProducts' are correlated with 'Income'.
- 'MntWines', 'MntMeatProducts', 'NumCatalogPurchases' are highly correlated with spending.
- 'MntMeatProducts', 'NumCatalogPurchases' are correlated.

	Age	Income	Spending	MntWines	MntFruits	MntMeatProducts	MntFishProducts	MntSweetProducts	MntGoldProds	NumDealsPurchases	NumWebPurchases	NumCatalogPurchases	NumStorePurchases	NumWebVisitsMonth
Age	1.00	0.20	0.11	0.16	0.01	0.03	0.04	0.02	0.06	0.07	0.15	0.13	0.14	-0.12
Income	0.20	1.00	0.79	0.69	0.50	0.68	0.52	0.52	0.38	-0.11	0.45	0.69	0.63	-0.64
Spending	0.11	0.79	1.00	0.89	0.61	0.84	0.64	0.60	0.52	-0.06	0.52	0.78	0.68	-0.50
MntWines	0.16	0.69	0.89	1.00	0.39	0.56	0.40	0.39	0.39	0.01	0.54	0.63	0.64	-0.32
MntFruits	0.01	0.50	0.61	0.39	1.00	0.54	0.59	0.57	0.39	-0.13	0.30	0.49	0.46	-0.42
MntMeatProducts	0.03	0.68	0.84	0.56	0.54	1.00	0.57	0.52	0.35	-0.12	0.29	0.72	0.48	-0.54
MntFishProducts	0.04	0.52	0.64	0.40	0.59	0.57	1.00	0.58	0.42	-0.14	0.29	0.53	0.46	-0.45
MntSweetProducts	0.02	0.52	0.60	0.39	0.57	0.52	0.58	1.00	0.37	-0.12	0.35	0.49	0.45	-0.42
MntGoldProds	0.06	0.38	0.52	0.39	0.39	0.35	0.42	0.37	1.00	0.05	0.42	0.44	0.38	-0.25
NumDealsPurchases	0.07	-0.11	-0.06	0.01	-0.13	-0.12	-0.14	-0.12	0.05	1.00	0.23	-0.01	0.07	0.35
NumWebPurchases	0.15	0.45	0.52	0.54	0.30	0.29	0.29	0.35	0.42	0.23	1.00	0.38	0.50	-0.06
NumCatalogPurchases	0.13	0.69	0.78	0.63	0.49	0.72	0.53	0.49	0.44	-0.01	0.38	1.00	0.52	-0.52
NumStorePurchases	0.14	0.63	0.68	0.64	0.46	0.48	0.46	0.45	0.38	0.07	0.50	0.52	1.00	-0.43
NumWebVisitsMonth	-0.12	-0.64	-0.50	-0.32	-0.42	-0.54	-0.45	-0.42	-0.25	0.35	-0.06	-0.52	-0.43	1.00



We will now test whether two samples has the same mean or not. (Test if Income of 'Australia', 'India' and 'Canada' has same mean)

For this we have two types of significance tests for two different conditions.

Parametric Statistical Significance Test

1.Student's t-test - It tests whether the two independent normal distributed samples has the same mean or not.
2.Analysis of Variance Test (ANOVA) - It tests whether the two or more independent normal distributed samples has the same mean or not. ANOVA is same as t-test but for more than 2 variables. So either we can apply t-test pair-wise of apply ANOVA once. Also ANOVA only tells whether all samples are same or not, it doesn't quantify which samples differ or by how much.

Non-Parametric Statistical Significance Test

1.Mann-Whitney U Test - Non-parametric equivalent of Student's t-test.
2.Kruskal-Wallis H - Non-parametric equivalent of ANOVA (it's for median).

We will apply appropriate test depending on the sample, i.e., if samples are normally distributed then parametric tests otherwise non-parametric tests.

Q1. Test if Income of 'Australia', 'India' and 'Canada' has same mean?

Customer income of Australia, India and Canada has the same mean.

Check for normal distribution

```
shapiro_wilk_test(df[df['Country']=='AUS'], ['Income'])
```

```
Fail to reject H0 due to lack of evidence under significance level 0.05
Income seem to be normally distributed
```

```
shapiro_wilk_test(df[df['Country']=='IND'], ['Income'])
```

```
Fail to reject H0 due to lack of evidence under significance level 0.05
Income seem to be normally distributed
```

```
shapiro_wilk_test(df[df['Country']=='CA'], ['Income'])
```

```
Fail to reject H0 due to lack of evidence under significance level 0.05
Income seem to be normally distributed
```

- The samples are normally distributed.
- Therefore apply ANOVA Test

H0 : Customer Income of Australia, India and Canada has same sample mean. α = 0.05

```
ALPHA = 0.05
```

```
_, p = stats.f_oneway(df[df['Country']=='AUS'].Income, df[df['Country']=='IND'].Income, df[df['Country']=='CA'].Income)

if p <= ALPHA:
    print(f'Rejected H0 under {ALPHA*100}% significance, Different distributions.')
else:
    print(f'Fail to Reject H0 under {ALPHA*100}% significance, Same distributions.')
```

```
Fail to Reject H0 under 5.0% significance, Same distributions.
```

Q2. Test whether wine purchase across Australia, India and Canada has the same mean?

Wine purchases across Australia, India and Canada are equal.

```
shapiro_wilk_test(df[df['Country']=='AUS'], ['MntWines'])
```

```
Rejected H0 under significance level 0.05
MntWines doesn't seems to be normally distributed
```

```
shapiro_wilk_test(df[df['Country']=='IND'], ['MntWines'])
```

```
Rejected H0 under significance level 0.05
MntWines doesn't seems to be normally distributed
```

```
shapiro_wilk_test(df[df['Country']=='CA'], ['MntWines'])
```

```
Rejected H0 under significance level 0.05
MntWines doesn't seems to be normally distributed
```

- The samples are not normally distributed.
- Therefore apply **Kruskal-Wallis Test** (non-parametric)

```
_, p = stats.kruskal(df[df.Country=='AUS'].MntWines, df[df.Country=='IND'].MntWines, df[df.Country=='CA'].MntWines)

if p <= ALPHA:
    print(f'Rejected H0 under {ALPHA*100}% significance, Different distributions.')
else:
    print(f'Fail to Reject H0 under {ALPHA*100}% significance, Same distributions.')
```

```
Fail to Reject H0 under 5.0% significance, Same distributions.
```

Q3. Is there association between Marital Situation and Income?

Solution:

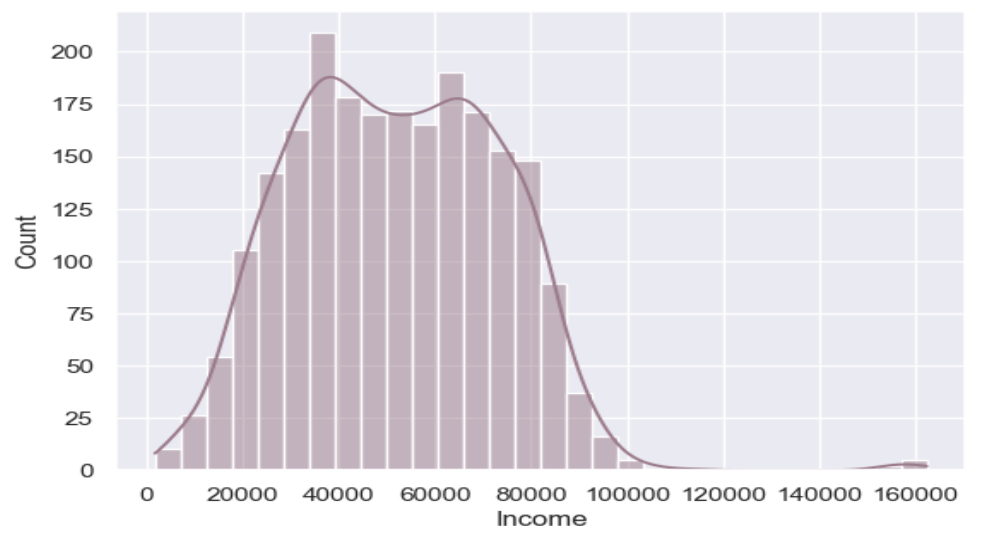
Point-Biserial Correlation
It is a special case of the Pearson Correlation and is used when you want to measure the relationship between a continuous variable and a dichotomous variable, or one that has two values (i.e. male/female, yes/no, true/false).

Assumptions for point biserial correlation:

No outliers (continuous variable) - assessed through a visual examination of the scatterplot etc
Approximately normally distributed (continuous variable)
Homogeneity of variance of the continuous variable between both groups of the dichotomous variable - assessed through Levene's Test

Outliers and Normality

- Income does not have extreme outliers.
- Income is normally distributed.



Q3. Is there association between Marital Situation and Income?

Check for Homogeneity: Levene's Test

- Thus variance among 'alone' and 'in couple' marital situation groups across Wine purchases is equal.
- Thus Assumption "homogeneity of variance of the continuous variable between both groups of the dichotomous variable" is verified
- Therefore apply point biserial

```
al_df = df[df['Marital_Situation']== 'Alone']['Income']
ic_df = df[df['Marital_Situation']== 'In couple']['Income']
```

The null hypothesis for Levene's test is that the variance among groups is equal

```
w_stats, p_value = levene(al_df,ic_df,
                          center='median')

if p_value > ALPHA:
    print("We do not reject the null hypothesis")
else:
    print("Reject the Null Hypothesis")
```

```
We do not reject the null hypothesis
```

The hypotheses for point biserial correlation:

Null hypothesis: The correlation coefficient r = 0 (There is no correlation)
Alternative hypothesis: The correlation coefficient r ≠ 0 (There is a correlation)

```
from scipy.stats import pointbiserialr
```

```
for col in ms:
    correlation, p_value = pointbiserialr(ms[col], df['Income'])
    print(f'{correlation:.4f}: Point-Biserial Correlation
                          for {col} with p-value {p_value:.4f}')
```

```
0.0034: Point-Biserial Correlation for Alone with p-value 0.8715
-0.0034: Point-Biserial Correlation for In couple with p-value 0.8715
```

- Income is normally distributed. P value greater than 0.05. Thus null hypothesis is not rejected.
- **No association between marital status and income.**

Q4. Is age group and number of Web visits per month independent?

Solution:

We need to test Hypothesis for Independence between two categorical variables

Pearson's Chi-square test

The Chi-square statistic is a non-parametric statistic tool designed to analyze group differences when the dependent variable is measured at a nominal level Assumptions:

- The test becomes invalid if any of the expected values are below 5
- The p value calculated is not exact but approximate and converges to exact on increasing data(so not good for small sample size)
- The number of observations must be 20+

```
observed_values = pd.crosstab(df['AgeGroup'], df['NumWebVisitsMonth'])
observed_values
```

NumWebVisitsMonth	0	1	2	3	4	5	6	7	8	9	10	13	14	17	19	20
AgeGroup																
Young	0	1	2	1	0	2	3	0	0	1	0	0	0	0	0	0
Young Adult	0	28	20	20	22	18	31	43	48	17	1	0	0	0	1	0
Adult	4	27	44	56	52	68	92	106	103	33	0	1	0	1	0	0
Senior	2	30	49	49	58	88	103	139	107	19	2	0	2	0	0	1
Retired	2	37	55	47	52	70	79	70	52	9	0	0	0	0	1	1
Old Age	3	29	32	32	33	34	31	35	32	4	0	0	0	0	0	1

Expected cell counts are small, it is better to use an exact test as the chi-squared test is no longer a good approximation in such cases.

Q4. Is age group and number of Web visits per month independent?

Fisher's exact test

- Fisher's exact test is used to determine whether there is a significant association between two categorical variables in a contingency table.
- Fisher's exact test is an alternative to Pearson's chi-squared test for independence.
- A general recommendation is to use Fisher's exact test- instead of the chi-squared test - whenever more than 20 % of cells in a contingency table have expected frequencies < 5.

```
chi2, p, dof, expected_values = stats.chi2_contingency(observed_values)
print(chi2, p, dof,"\n", expected_values)
```

```
134.30750457213958 3.135219316047695e-05 75
 [[4.91949911e-02 6.79785331e-01 9.03398927e-01 9.16815742e-01
  9.70483005e-01 1.25223614e+00 1.51610018e+00 1.75760286e+00
  1.52951699e+00 3.71198569e-01 1.34168157e-02 4.47227191e-03
  8.94454383e-03 4.47227191e-03 8.94454383e-03 1.34168157e-02]
 [1.22495528e+00 1.69266547e+01 2.24946333e+01 2.28287120e+01
  2.41650268e+01 3.11806798e+01 3.77508945e+01 4.37643113e+01
  3.80849732e+01 9.24284436e+00 3.34078712e-01 1.11359571e-01
  2.22719141e-01 1.11359571e-01 2.22719141e-01 3.34078712e-01]
```

```
if p <= ALPHA:
    print(f'Rejected H0 under significance level {ALPHA} `Age` & `Web Visits Per Month` are dependent.')
else:
    print(f'Fail to reject H0 due to lack of evidence under significance level {ALPHA} `` & `Country` are independent.')
```

```
Rejected H0 under significance level 0.05 `Age` & `Web Visits Per Month` are dependent.
```

- **Thus, Number of Web Visits Per Month are dependent on Age of the customers.**
- **It is evident that different generations have different intensity of using technology.**